Dr. Dandamudi Usharani



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1. **Fields of Specialization**: Quality Assurance of Oils & Fats, Pesticides, Chemical Biology, Computational Bioinorganic Chemistry, Lipidomics, Molecular Modeling

2. **Education**

Ph.D.	2009	Univ. of Hyderabad, India	Applied Theoretical Chem.
M.Sc.	1996	Univ. of Hyderabad, India	General Chemistry

3. **Research Experience**: **20 years** (5 years in Israel)
- Quality Assurance & Lipidomics CSIR-CFTRI, India, 2017-current
- Lipidomics & Molecular Modeling (Quick Hire Scientist), CSIR-CFTRI, India 2014-2017
- Computational Chemistry (Post doc) The Hebrew Univ. of Jerusalem, Israel 2009-2014
- Applied Theoretical Chemistry (Res. Fellow) Univ of Hyderabad, India 2004-2009

4. Teaching Experience: 15 years

M. Sc Bioinformatics, Chemical Sciences, Good Laboratory Practices, Biotechniques, Food Safety at CSIR-CFTRI, 9 years
B.Sc. Chemistry, MNR Degree College, Hyd. 2 years
Intermediate Chemistry Guntur Vikas Junior College, & Vignan Junior College, Hyd. 4 years

5. **Scientific Activities:**
- Member of American Chemical Society, Chemical Research Society of India
- Member of World Association of Theoretical and Computational Chemists
- Member of Society of Biological Chemists (India), Indian Science Congress
- Member of Indian Science Congress Association, Member of The Association of Analytical Communities (AOAC) International, India
- Student Committee Member of Association of Food Scientist & Technologists (India)
- Member of Scientific Panel Pesticides, Served as Member of Scientific Panel of Oils and Fats
- Member of Bureau of Indian Standards Sectional Committee of Chemical Hazards (CHD 07), Oils & Oil Seeds (FAD 13), Food Hygiene, Safety Management and other systems (FAD 15) and Test methods for Food Products (FAD 28),
- Served as Research Council Member for ICAR-NISA, Ranchi 2020-2023
- Served as Member of 'Network for Scientific Co-operation for Food Safety and Applied Nutrition' for Foods of Plant Origin, Scientific Panel of Oils and Fats. Exam paper setting for Agricultural scientists recruitment board (ASRB)
- Served as Expert Member to WHO for Simplified protocol for Measuring Trans Fatty acids in Foods.
- Member of EWG of Pesticides CCPR and Chair for Electronic Working group for Discussion paper of Acrylamide in foods CCCF17 and CCCF18.

6. **Honors and Awards:**
- **Mrs Vaidehi Kishan Rao Memorial Gold Medal** from Osmania University, B.Sc., 1996
- Qualified CSIR-NET (National Eligibility Test) (2003); CSIR-JRF & SRF (2004-2009)
- Best poster awards in **ACS-CSIR-AOCCB conference-2006** and in **Theoretical Chemistry Symposium-2009** and in **Current Trends in Theoretical Chemistry-2013,** Best poster award in

Society of Biological Chemists (2017), International Conference on Bioactive Lipids in Health and Diseases, 2024.

7. **Research supervision**:	Ph. D Awarded	**4**	Govt. funded projects	**10**
	M. Sc. students	**22**	Institutional grants	**3**
	Project Associates	**9**	Industrial grants	**5**
	Research Fellows	**2**	Research Visitors	**2**

Granted Projects **21**

- Industrial Sponsored project from Vyome Biosciences Pvt. Ltd, New Delhi, India ***6.58*** *lakh* SSP 210, (PI) Mar-2015 - Jun-2015
- Industrial Consultancy project from Vyome Biosciences Pvt. Ltd, New Delhi, India ***6 lakhs***, CNP 523 (PI) Jun 2016 - Jun 2017
- CSIR Project: National Centre for Advanced Analytical Services Theme: Agriculture, Nutrition and Biotechnology, ***100 lakhs*** MLP 232 (Member) Apr 2018 - Mar 2019
- National Referral Laboratory, FSSAI India ***50 lakhs***,GAP 559 (Member) Aug2018 Aug2021
- Capacity Building of State of Art Analytical Facilities to Institute of Food Security, FCI, India, 25 lakhs CNP 536 (Member) Apr 2017 Apr 2021
- CSIR FOCUS project on Understanding structure function relationships in enzymes critical for the survival of bacterial food pathogens Theme: Agriculture, Nutrition and Biotechnology, CSIR, India ***80 lakhs***, MLP 275 (Co-PI) May 2019 - May 2023
- Network for scientific cooperation for food safety and applied nutrition., FSSAI India ***50 lakhs***, GAP 575 (Member) Apr 2019 Apr 2023
- Industrial Sponsored M/s Anagha Refineries Pvt, Ltd, Mangalore, India ***10 lakhs*** SSP 283 (Member) Jun 2020 Dec 2021
- DST-SERB Accelerate Vigyan Karyashala Workshop Hands-on Training and Workshop on High End Mass Spectrometry Based Lipidomics for Food and Biomedical Sciences **4 lakhs** TSP 52 (PI) Nov 2022- Mar 2023
- ICMR Adhoc: Prevention of Cardiometabolic Risk Factors by Multi-Sourced Edible Oil Fortified with Lipid Nutraceuticals: Nutritional Intervention for Obesity related Complications. ***35 Lakhs*** Apr 2022 Apr 2025
- DST-SERB: Probing Structure and Reaction Mechanisms of Microsomal Epoxide Hydrolases through multi scale modeling to understand lipid signalling and metabolism ***24.6 lakhs*** GAP 621 Jul 2022 - Jul 2025
- Industry Sponsored project Analytical characterization of A Hango related products ***3.96 lakhs*** SSP 346 (PI) Feb 2023 Aug 2023
- FSSAI funded Training and Capacity Building Programme for Food Testing Laboratories 56 Lakhs GAP 639 (Member) Aug 2023 Mar 2024
- CSIR Millet Mission Global Awareness of Health and Nutritional Attributes Of millets HCP 52 (Co-PI) ***102 Lakhs*** Aug 2023 Mar 2025
- IIT-Delhi- Bill Melinda Gate Foundation Analytical Quality Control and Quality Assurance in Rice Fortification ***204 Lakhs*** GAP 640 (Member) Aug 2023 Aug 2026
- Bureau of Indian Standards -Validation of published test methods of Vitamins in identified food matrices for Revision of Indian Standards ***8 Lakhs*** GAP 644 (PI) Dec 2023 Aug 2024
- Karnataka-Health Promotion Trust- Quantification of Vitamin A and D in Oils and Cooking Conditions **25.6 Lakhs** SSP 359 (PI) Jun 2024 July 2025
- FSSAI- Validation of Analytical Methods for Precise Quantification of Isoflavones, Anthocyanins, Lycopene, Lutein and Astaxanthin and Chondroitin Sulfate and Hyaluronic Acid in Functional Foods ***65.7 Lakhs*** GAP 653 (Nodal officer) Oct 2024 Oct 2026
- Setting up the State of Art Laboratory for Quality and Safety Assurance of Agricultural and value added Products **27 Lakhs**, CNP 576 Oct 2025 Oct 2026
- FSSAI funded Setting up of Organic Food Testing Facility **197.8 Lakhs** Feb 2026 March 2030
- Accelerating trans fat elimination in India to reduce cardiovascular diseases **55.28 Lakhs** GAP 666 (PI) Feb 2026- Feb 2027

8. Research Publications

Original Scientific articles published *(foreign journals)* ***60***
Reviews + Book Chapters ***3+ 5+2***
Citations (Google Scholar) ***2111 h-index*** (Google Scholar) ***20*** ORCID: **0000-0001-5728-9421**

9. Publications in a nutshell 60

Authors	*Journal*	*Year*	*Vol.*	*Pages*
60. A. Bakshi, D. Usharani*	**Comp. Biol. and Med**.	2025	119,	111279.
59. Swapna… D. Usharani, Gokulkrishnan	**Cardiovascular Diabetology**	2025	24	434
58. S. Kottekad, D. Usharani,*	**Plant Foods Hum. Nutr**	2025	80	65
57. K. S. Naresh, D. Usharani, S.R. Chaudhari	**Food Hydrocolloids**	2025	162	110958
56. P. Kumari, D. Usharani, Ravi Kumar	**Biochimie**	*2024*	*i14,*	*S0300*
55. S. Sharma, Y.V. Subrahmanyan…	**Nutr. Met. Card. Dis.,**	2024,	24,	1807
54. M Jana,….. D Usharani, * H Rath	**Dalton, Trans.,**	2024,	55,	7397
53. R Narayanasamy, D Usharani, R Rajasekharan,	**Chem. Phys. Lipids**	2024,	258,	105353
52. K Sanjay, R Sudip, D Usharani	**J. Biomol. Stru. Dyn.,**	2024,	, 42,	3472
51. R Narayanasamy, R Rajasekharan, D Usharani	**Biophys. Chem.,**	2023,	296,	106976
50. S. Sahoo,…D. Usharani,* H. Rath*,	**Organic Lett.,**	2023,	25,	1491
49. S. Sahoo,…D. Usharani,* H. Rath*,	**J. Porhyr. Phthalocyan**	2023,	(01n04),	82
48. N. Halder,…D. Usharani,* H. Rath*, M. Jana	**Chem. Asian. J.,**	2022,	17	e202200108
47. N. Halder, R…. D. Usharani,* H. Rath*	**Org. Chem. Front**.,	2022,	9	2333
46. L. Anusree….. D. Usharani, Ravi Kumar	**BBA Proteomics**.,	2022,	1870,	140781
45. M. Jana….. D. Usharani*, H. Rath*	**Org. Biomol. Chem**.,	2022,	20,	6741
44. D. Usharani*, ---- S. Fornarini	**Int. J. Mass. Spec.**	2022	471	116724
43. B. Chakraborty… D. Usharani, H Rath	**Dalton Trans.**	2021	50	*14421*
42. S. Murugesan D.Usharani, S.N Mudliar	***Comput. Biol. Med***	2021	136	104683
41. N. Halder---- D. Usharani,* H. Rath	***J. Porphyr. Phthalocyan***	2021	25	966
40. N. Halder---- D. Usharani,* H. Rath	***J. Org. Chem.***	2021	86	8015
39. N. Halder---- D. Usharani,* H. Rath	***J. Porphyr. Phthalocyan***	2021	25	436
38. L. Anusree .... D. Usharani, Ravi Kumar	***Biochimie***	2021	182	217
37. S. Sahoo----D. Usharani, H. Rath	***Org. Biomol. Chem***	2020	18	6058
36. S. Abduldileep--D. Usharani,- R. Rajasekharan	***Int. J Biol. Macromo*l.**	2020	156	938
35. G. S. Kaira. D. Usharani, M. Kapoor	***Enzy. Micro. Tech***.	2020	135	109497
34. N. Halder----D. Usharani, H. Rath	***J. Org. Chem.***	2020	85	2059
33. K. C. Sahoo----D. Usharani, H. Rath	***J. Porphyr. Phthalocyan***	2020	24	161
32. N. Halder----D. Usharani, H. Rath	***Org. Biomol. Chem***	2019	17	6131
31. K. C. Sahoo, D. Usharani, H. Rath	***J. Org. Chem.***	2019	84	5203
30. G. S. Kaira, D. Usharani, M. Kapoor	***Int. J Biol. Macromo*l.**	2019	133	1236
29. J. Anusha, D. Usharani--- R. Rajasekharan	***BBRC*.**	2018	507	155
28. S Ghosh---D. Usharani, S. Sengupta	***J. Investig. Dermatol.***	2018	138	1400
27. D. Dutta---D. Usharani, S. Mazumdar	***Dalton Trans***	2016	45	17624
26. J. Li---D. Usharani---H. Schwartz	***J. Am. Chem. Soc*.**	2016	138	11368
25. J. Li---D. Usharani----H. Schwartz	***J. Am. Chem. Soc*.**	2016	138	7973
24. S. Shaik, D. Usharani	**Book John Wiley & Sons**	ISBN: 978-1-119-00140-9		
23. D. Usharani B. Wang… S. Shaik, **Book Chapter John Wiley & Sons.**		ISBN:978-1-118-89831-4.		
22. D. Mandal---D. Usharani..S. Shaik	***J. Am. Chem. Soc.***	2015	137	722
21. B. Wang----D. Usharani..S. Shaik	***J. Am. Chem. Soc.***	2014	136	13895
20. D. Usharani--- S. Shaik	***Chem. Soc. Rev.***	2014	43	4968
19. Y. Yang---- D. Usharani--- L Waskell,	***Biochemistry***	2014	53	5080
18. D. Usharani----S. Shaik,	***J. Am. Chem. Soc.***	2013	135	17090
17. S. Shaik---- D. Usharani, W. Thiel, **Book Chapter John Wiley & Sons.**		ISBN:978-3-527-33566-4		
16. D. Usharani----S. Shaik	***Acc. Chem. Res.***	2013	46	471
15. Y. Wang----D. Usharani, S. Shaik	***ACS Catalysis***	2013	3	1334
14. R. Mas-Ballesté D. Usharani-- S. Shaik	***Chem.− Eur. J***.,	2012	18	11747
13. D. Usharani--- S. Shaik	***J. Am. Chem. Soc.***	2012	134	4053
12. D. Janardanan, D. Usharani, S. Shaik	***Angew. Chem. Int. Ed*.,**	2012	51	4421
11. D. Usharani--- S.Shaik	***J. Am. Chem. Soc.***	2011	133	176
10. S. Shaik--- D. Usharani	***J. Chem. Theory and Comp***	2011	7	327
9.D. Janardanan, D. Usharani----S. Shaik	***J. Phys. Chem. Lett***	2011	2	2610
8.P. Schymann, D. Usharani---- S. Shaik	***J. Phys. Chem. B***	2010	114	7078

7. K.H.K.Reddy, D. Usharani---E. D. Jemmis	***Chem.− Eur. J***.	2011	17	9142
6. D. Usharani---- E. D. Jemmis	***Chem.− Eur. J***.,	2009	15	8429
5. S. Ghosh, D. Usharani---- E. D. Jemmis	***Chem. An Asian J***	2008	3	1949
4. S. Ghosh, D. Usharani---- E. D. Jemmis	***Bioconj. Chem***	2008	19	2332
3. P. Srivani, D. Usharani--- G.N. Sastry	***Curr. Pharma. Design****.,*	2008	14	3854
2. D. Usharani------ E. D. Jemmis	***J. Chem. Theory and Comp***	2008	4	974
1. D. Usharani, DLVK Prasad, E. D. Jemmis,	***J. Comput. Chem***	2007	28	310

Professional Experience

Dr. Dandamudi Usharani, Senior Principal Scientist and Group Leader of the Oils and Fats Division at the Department of Food Safety and Analytical Quality Control Laboratory, CSIR-CFTRI, has made significant strides in food safety, with a focus on oils and fats chemistry. As a leader in her field, she has developed cutting-edge in-house mass spectrometric and chromatographic techniques for the quantification of antioxidants, vitamins, fatty acid profiling, and bioactive lipids. These innovations have greatly enhanced the precision and accuracy of lipidomics, setting higher standards for safety and quality in food products and lipid metabolism.

Dr. Usharani's research also delves into molecular modeling, where she explores the electronic structure, bonding, and spectroscopic characteristics of fats and oils. Her work on bioactive lipid species, catalytic compounds, and organometallic complexes provides a deeper understanding of their underlying chemistry for molecular mechanisms. She is also actively involved in studying protein-ligand interactions, which are crucial for rationalizing mechanisms and designing selectivity in Structure-Based Drug Design (SBDD). This work advances knowledge of the structure-function relationships of biomolecules and bioactive compounds in food, and their role in disease mechanisms. Her research has resulted in 60 publications in international peer-reviewed journals and eight book chapters and a book. In addition to her research, Dr. Usharani plays a key role in collaborative projects with industry and government bodies aimed at improving food safety regulations. Her leadership at CFTRI has driven significant advancements in analytical quality control, ensuring that food products meet stringent safety and quality standards. She also provides scientific input to both national and international regulatory bodies. Dr. Usharani serves on the Scientific Panel of FSSAI, the Bureau of Indian Standards, and is an Expert Member on Trans Fatty Acids for World Health Organization (WHO) as well as the Electronic Working Group on Codex Contaminants in Foods.